

Random Matrices and Gaussian β -Ensembles

A **Random Matrix** is a matrix where all or some of the entries are random variables drawn from a specific probability distribution. In our project, we focus on the limiting behavior of the eigenvalues of the matrix.

Our project focuses on a one-parameter family of models called the **Gaussian β -Ensembles** which have the density function given by

$$p_{\beta}(\lambda_1, \lambda_2, \dots, \lambda_N) = \frac{1}{Z_{N,\beta}} \prod_{1 \leq i < j \leq N} |\lambda_i - \lambda_j|^{\beta} e^{-\frac{\beta}{4} \sum_{i=1}^N \lambda_i^2},$$

where $Z_{N,\beta}$ is a normalizing constant. When $\beta = 1, 2, 4$, this corresponds to a random matrix with real, complex, and quaternion standard normal elements. These are the classical cases, but we can generalize the model for any $\beta \in \mathbb{R}_{>0}$. In our project, we focus on the situation when $\beta = 2n$, $n \in \mathbb{N}$. The histograms of the eigenvalues satisfy a famous result—*Wigner's Semicircle Law*.

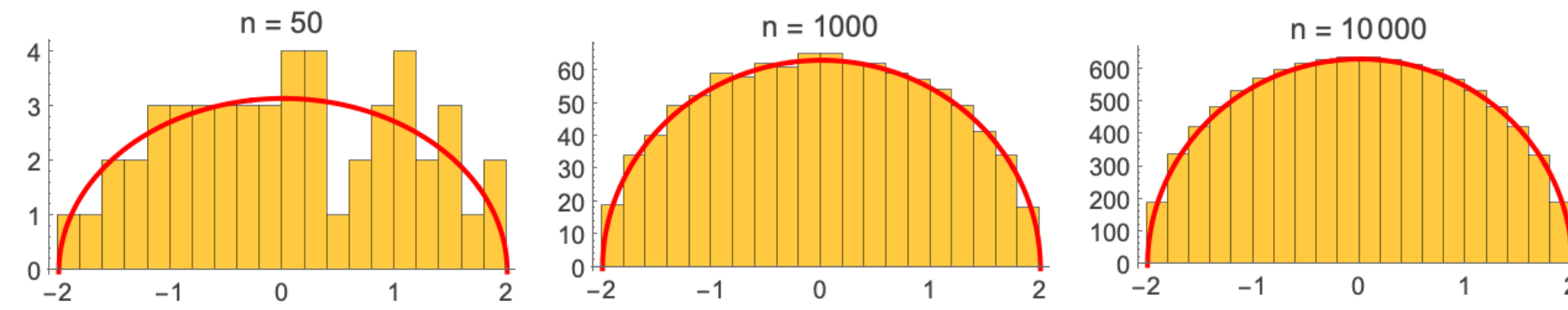


Fig. 1: Simulation of Wigner's Semicircle Law

Point Processes and Correlation Function

As $N \rightarrow \infty$, eigenvalues in the interior (bulk) form a limiting **point process**—a random collection of points. For $\beta > 0$, this is known as the Sine_{β} process.

The **pair correlation function** $\rho_{\beta}^{(2)}(0, \lambda)$ gives a measure of the likelihood of finding 2 particles with distance λ apart from each other. Qu and Valkó [2] proved the function is given by an infinite series in terms of a random variable $\alpha\lambda$:

$$\rho_{\beta}^{(2)}(0, \lambda) = \frac{1}{4\pi^2} + \frac{1}{2\pi^2} \sum_{k=1}^{\infty} \frac{(-\beta/2)^{\dagger k}}{(1 + \beta/2)^{\dagger k}} E[\cos(k\alpha\lambda)]$$

When $\beta = 2n$, the *rising factorial* $(x)^{\dagger k} = x(x+1)(x+2) \cdots (x+k-1)$ truncates the series into a finite sum because $(-n)^{\dagger n+1} = 0$.

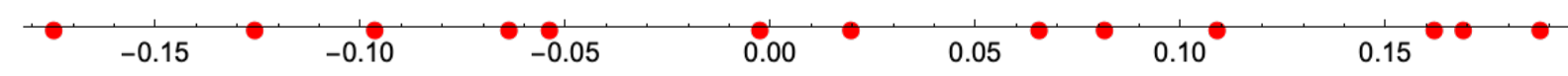


Fig. 2: Simulation of a Point Process for $\beta = 1$

Forrester Integral Formula

When $\beta = 2n$, Peter Forrester showed there is an integral formula for the 2-point correlation function [1]:

$$\rho_{\beta=2n}^{(2)}(0, \lambda) = \frac{1}{4\pi^2} \cdot \frac{n^{2n}(n!)^3}{(2n)!(3n)!S_{2n}(-1+1/n, -1+1/n, 1/n)} \times \int_{[0,1]^{2n}} \prod_{j=1}^{2n} \left(e^{i\lambda u_j} u_j^{-1+1/n} (1-u_j)^{-1+1/n} \right) \prod_{j < k} |u_j - u_k|^{2/n} \prod_{j=1}^{2n} du_j,$$

where $S_{2n}(-1+1/n, -1+1/n, 1/n)$ is defined by

$$\int_{[0,1]^{2n}} \prod_{j=1}^{2n} \left(u_j^{-1+1/n} (1-u_j)^{-1+1/n} \right) \prod_{j < k} |u_j - u_k|^{2/n} \prod_{j=1}^{2n} du_j.$$

By applying the change of variables $u_j = t_j + \frac{1}{2}$, and the Euler's Formula, we showed that the above integral is **indeed real**.

The Vector-Valued ODE System

For $\beta = 2n$, we define a vector-valued function $\mathbf{q}(\lambda) = [q_1(\lambda), \dots, q_n(\lambda)]^T$ where $q_k(\lambda) = E[e^{ik\alpha\lambda}]$. [2] proved that this vector obeys the following first-order linear ODE system:

$$\frac{\beta}{4} \lambda \mathbf{q}'(\lambda) = \left(i \frac{\beta}{4} \lambda \mathbf{B}_n + \mathbf{A}_n \right) \mathbf{q}(\lambda) + \frac{n+1}{2} \mathbf{e}_n,$$

Where $\mathbf{A}_n$ is a tri-diagonal and $\mathbf{B}_n$ is a diagonal matrix with an explicit formula. The correlation function is cleanly recovered via the real part of $\mathbf{q}$:

$$\rho_{2n}^{(2)}(0, \lambda) = \frac{1}{4\pi^2} (1 + 2\mathbf{v}_n^T \Re \mathbf{q}(\lambda)), \quad (1)$$

where $[\mathbf{v}_n]_k = (-1)^k \binom{2n}{n+k} / \binom{2n}{n}$, $1 \leq k \leq n$.

Algorithm for ODE System Reduction

We implemented a Mathematica algorithm to automate the reduction of the n -dim system into a single variable $2n$ -order ODE with for $h(\lambda) := \Re \mathbf{v}_n^T \mathbf{q}(\lambda)$:

- **Substitution Rules:** Express $\Re q'_k$ and $\Im q'_k$ as linear combinations of $\Re \mathbf{q}$ and $\Im \mathbf{q}$.
- **Recursive Reduction:** Repeatedly differentiate $h(\lambda) = \Re \mathbf{v}_n^T \mathbf{q}(\lambda)$ and substitute lower-order terms to eliminate variables.
- **Reduction to ODE system:** h together with the derivatives $h^{(1)}, \dots, h^{(2n-1)}$ gives a system of ODE with respect to $\Re \mathbf{q}, \Im \mathbf{q}$.
- **Recover the ode for h :** plugging back the solutions of $\Re \mathbf{q}, \Im \mathbf{q}$ in terms of $\{h^{(k)}\}_{k=0}^{2n-1}$ to the $2n^{th}$ derivative $h^{(2n)}$ of h yields a $2n$ -order ODE for $\rho_{2n}^{(2)}$ for any even β .

```
In[2043]:=
twoPointCorr[3]
twoPointCorr[4]
twoPointCorr[5]

Out[2043]=
-1/2 + 2 \pi^2 rho[\lambda] == -1/12 (14 - 33 \lambda^2 + 18 \lambda^4 + 27 \lambda^6)
(96 \pi^2 \lambda (-11 - 3 \lambda^2 + 27 \lambda^4) rho'[\lambda] + 6 \pi^2 \lambda^2 (-268 - 732 \lambda^2 + 147 \lambda^4) rho''[\lambda] +
2 (42 - 99 \lambda^2 + 54 \lambda^4 + \lambda^3 (2 \pi^2 (520 + 588 \lambda^2) rho^{(3)}[\lambda] +
7 \lambda (2 \pi^2 (62 + 9 \lambda^2) rho^{(4)}[\lambda] + 24 \pi^2 \lambda rho^{(5)}[\lambda]))) + 18 \pi^2 \lambda^6 rho^{(6)}[\lambda])

Out[2044]=
1/2 + 2 \pi^2 rho[\lambda] == -1/4 (-625 + 2704 \lambda^2 + 384 \lambda^4 (-7 + 3 \lambda^2 + 6 \lambda^4))
(-1250 + 32 \lambda^2 (169 - 168 \lambda^2 + 72 \lambda^4) + 8 \pi^2 \lambda (3271 + 32 \lambda^2 (-99 - 104 \lambda^2 + 288 \lambda^4)) rho'[\lambda] +
4 \pi^2 \lambda^2 (2831 - 25924 \lambda^2 + 33504 \lambda^4 + 6560 \lambda^6) rho''[\lambda] +
\lambda^3 (2 \pi^2 (-31954 + 40248 \lambda^2 + 39360 \lambda^4) rho^{(3)}[\lambda] +
\lambda (2 \pi^2 (429 + 42600 \lambda^2 + 4368 \lambda^4) rho^{(4)}[\lambda] + 8 \lambda (8 \pi^2 (440 + 273 \lambda^2) rho^{(5)}[\lambda] +
\lambda (2 \pi^2 (567 + 60 \lambda^2) rho^{(6)}[\lambda] + 2 \lambda (60 \pi^2 rho^{(7)}[\lambda] + 2 \pi^2 \lambda rho^{(8)}[\lambda])))

Out[2045]=
1/2 + 2 \pi^2 rho[\lambda] ==
-((16032016 + 200 \lambda^2 (-555419 + 25 \lambda^2 (35041 - 23925 \lambda^2 + 9000 \lambda^4)) + 320 \pi^2 \lambda
(-1471922 + 5 \lambda^2 (670718 + 5 \lambda^2 (-25611 + 50 \lambda^2 (-2389 + 4500 \lambda^2)))) rho'[\lambda] + 8 \pi^2 \lambda^2
(18304396 + 25 \lambda^2 (8086704 + 5 \lambda^2 (-3656523 + 3380700 \lambda^2 + 658625 \lambda^4))) rho''[\lambda] +
\lambda^3 (32 \pi^2 (36932444 + 25 \lambda^2 (-4048256 + 2664265 \lambda^2 + 2634500 \lambda^4)) rho^{(3)}[\lambda] +
\lambda (2 \pi^2 (-319693104 + 25 \lambda^2 (-7497896 + 53959950 \lambda^2 + 4778125 \lambda^4)) rho^{(4)}[\lambda] +
5 \lambda (8 \pi^2 (-12249496 + 275 \lambda^2 (124208 + 52125 \lambda^2)) rho^{(5)}[\lambda] + 55
\lambda (2 \pi^2 (326444 + 859500 \lambda^2 + 58125 \lambda^4) rho^{(6)}[\lambda] +
5 \lambda (16 \pi^2 (5554 + 2325 \lambda^2) rho^{(7)}[\lambda] + 5 \lambda (2 \pi^2 (1506 + 125 \lambda^2)
rho^{(8)}[\lambda] + 200 \pi^2 \lambda rho^{(9)}[\lambda])) + 6250 \pi^2 \lambda^5 rho^{(10)}[\lambda]))) /
(32064032 + 400 \lambda^2 (-555419 + 25 \lambda^2 (35041 + 75 \lambda^2 (-319 + 60 \lambda^2 (2 + 5 \lambda^2))))))
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Fig. 3: Mathematica Program Output

To verify our generated ODEs, we compute the series expansion $\mathbf{q}(\lambda) = \sum s_j \lambda^j$. The coefficients s_k follow the recursion:

$$s_k = i(kI - \frac{4}{\beta} A_n)^{-1} B_n s_{k-1}$$

The numerical consistency between the series and ODE solutions confirms our algebraic derivations.

Classical Cases: $\beta = 2$ and $\beta = 4$

- When $\beta = 2$ ($n = 1$): the system collapses to a single *1st-order scalar ODE*:

$$\frac{1}{2} \lambda q_1'(\lambda) = \left(\frac{i}{2} \lambda - 1 \right) q_1(\lambda) + 1, \quad q_1(0) = 1$$

Solving this yields the famous **Sine kernel**:

$$\rho_2^{(2)}(0, \lambda) = \frac{1}{4\pi^2} \left[1 - \left(\frac{\sin(\lambda/2)}{\lambda/2} \right)^2 \right]$$

- When $\beta = 4$ ($n = 2$): In this case we have a 2^{nd} -order ode by eliminating q_1 yields a *2nd-order ODE* for q_2 :

$$\lambda^2 q_2''(\lambda) = (-6\lambda + 3i\lambda^2) q_2'(\lambda) + (2\lambda^2 + 8i\lambda - 6) q_2(\lambda) + 6$$

Solving this correctly recovers the classical **Pfaffian correlation** for the Symplectic ensemble:

$$\rho_4^{(2)}(0, \lambda) = \frac{1}{4\pi^2} \left[1 - \text{sinc}^2(\lambda) + \text{sinc}'(\lambda) \int_0^\lambda \text{sinc}(t) dt \right]$$

$\beta = 6$ ODE and Its Solution

For $\beta = 6$, we derived the 3rd-order complex ODE $L(q_3) = h$ for q_3 , where L is 3rd order differential operator with polynomial coefficients, and h is explicit.

Operator Factorization: Using a heuristic search, we find that the 3rd-order differential operator L can be factored into a composition of a 1st-order operator P_1 and a 2nd-order operator P_2 :

$$L = P_1 \circ P_2$$

By repetitively differentiating the operator L and plugging $\lambda = 0$, we find that the boundary condition for the ODE of q_3 are $q_3(0) = 1$, $q_3'(0) = i$, and $q_3''(0) = -\frac{9}{8}$. Using the boundary conditions and solving the first and second order ODEs one after another we get a formula for q_3 . Using (1) we get an expression for $\rho_6^{(2)}$ which seems to be new.

Future Work

We aim to further understand the $\beta = 6$ case by investigating why the third-order ODE operator for q_3 admits a factorization. We also aim to study the $\beta = 8$ case and explore whether a similar operator decomposition can be derived there.

References

References

- [1] Peter J. Forrester. *Log-Gases and Random Matrices*. Princeton University Press, 2010.
- [2] Yahui Qu and Benedek Valkó. On the pair correlation function of the sine_{β} process. *arXiv preprint arXiv:2509.15446*, 2025.